

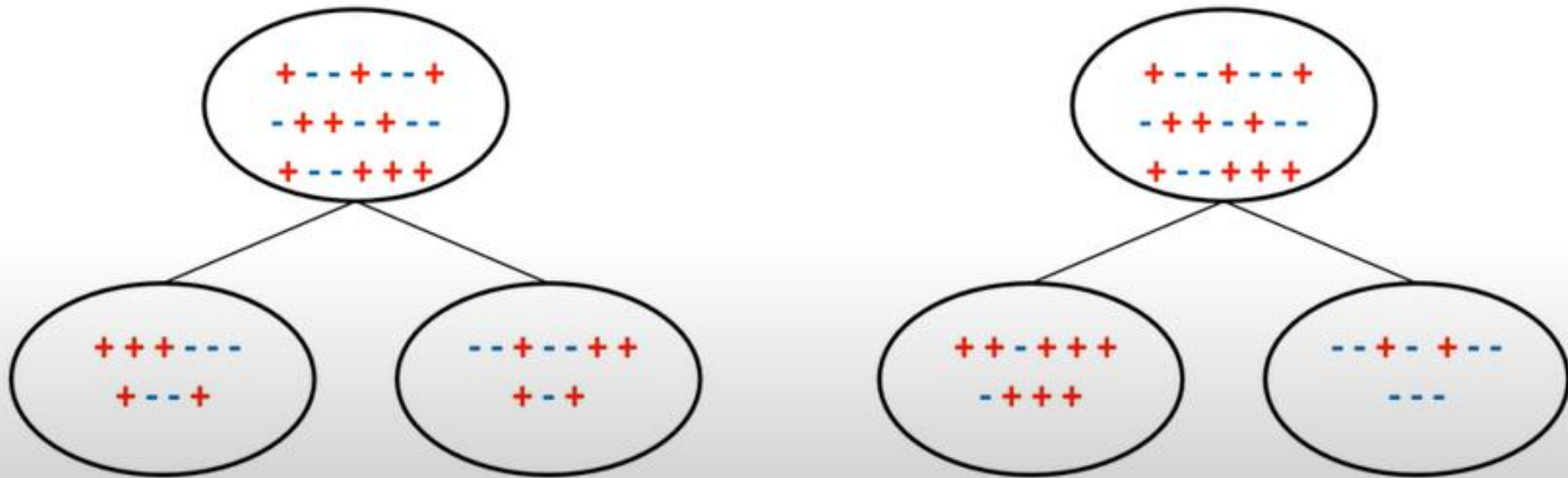
Machine Learning

Instructor: Hafiz Abdul Rehman

Lecture 23: Decision Tree – ID3

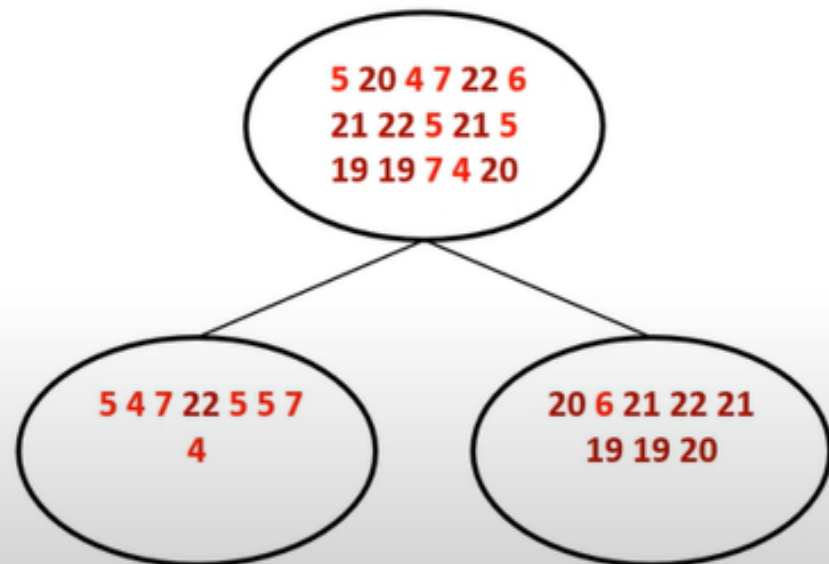
How do we grow the tree automatically

- We have seen pre-grown trees and have some ideas of manually creating them by trial and error.
- Is there a more systematic, quantifiable approach towards choice of feature for creating the splits and growing the tree step-by-step.
- Which one of the following is the better split?
 - Why?



How do we grow the tree automatically

- We have seen pre-grown trees and have some ideas of manually creating them by trial and error.
- Is there a more systematic, quantifiable approach towards choice of feature for creating the splits and growing the tree step-by-step.
- Which one of the following is the better split?
 - Why?



How do we grow the tree automatically

- We have seen pre-grown trees and have some ideas of manually creating them by trail and error.
- Is there a more systematic, quantifiable approach towards choice of feature for creating the splits and growing the tree step-by-step.

We prefer splits that lead to “**pure**” outcomes.
We need measure for “**purity**” and “**impurity**”



Training a Decision Tree – The ID3 and CaRT

Iterative Dichotomizer (ID3), Classification and Regression Trees

Split (node, {instances}, {attributes})

1. $A \leftarrow$ the best attribute for splitting the examples
 - In case of real-valued attributes, scan across all thresholds ($t \in T$) for each attribute
 2. Decision attribute for this node $\leftarrow A$ (threshold $\leftarrow T$)
 3. For each value of A , create a new child node
 - Create two children around the threshold for real-valued attributes
 4. Split training examples among child nodes
 5. For each child node/subset
 1. If subset is pure (or some other stopping criterion is met), STOP
 2. Else: split(child node, {subset of instances})
- Ross Quinlan, CS (ID3: 1986, C4.5: 1993)
 - Uses entropy as the impurity function
 - Meant primarily for categorical attributes and for classification
 - Breiman et al, Statistics (CaRT: 1984)
 - Uses Gini impurity
 - Meant for classification and regression

Measuring Impurity

Data: $S = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, $y_i \in \{1, \dots, c\}$, where c is the number of classes

- Let $S_k \subseteq S$, where $S_k = \{(x, y) \in S : y = k\}$ i.e. all inputs with label k .
- $S = S_1 \cup \dots \cup S_c$
- $p_k = \frac{|S_k|}{|S|}$, fraction of inputs in S with label k

Classification

- Gini Impurity

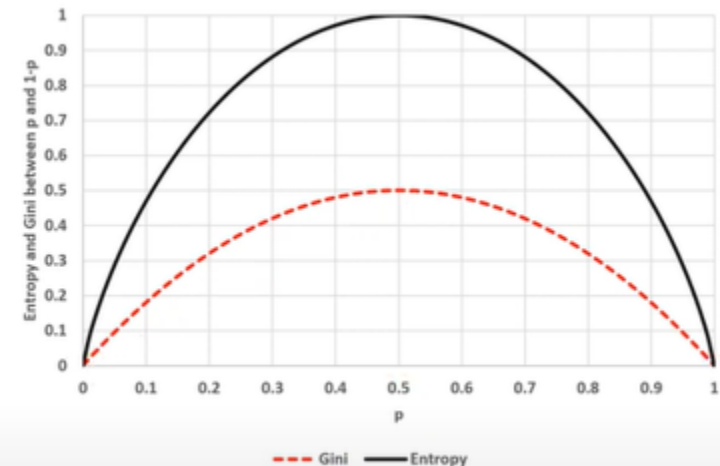
$$G(S) = \sum_{k=1}^c p_k(1 - p_k)$$

- Entropy

$$H(S) = \sum_{k=1}^c p_k \log_2\left(\frac{1}{p_k}\right)$$

$$H(S) = \sum_{k=1}^c p_k [\log_2(1) - \log_2(p_k)]$$

$$H(S) = - \sum_{k=1}^c p_k \log_2(p_k)$$



Gini and Entropy

$$G(S) = \sum_{k=1}^c p_k(1 - p_k),$$

$$H(S) = \sum_{k=1}^c p_k \log_2\left(\frac{1}{p_k}\right)$$

$$p_+ = \frac{10}{11} = 0.90, \quad p_- = \frac{1}{11} = 0.09, \quad Gini = 0.16, \quad H = 0.43$$



$$p_+ = \frac{20}{21} = 0.95, \quad p_- = \frac{1}{21} = 0.047, \quad Gini = 0.09, \quad H = 0.28$$



$$p_+ = \frac{7}{14} = 0.5, \quad p_- = \frac{7}{14} = 0.5, \quad Gini = 0.5, \quad H = 1.0$$



$$p_+ = \frac{10}{20} = 0.5, \quad p_- = \frac{10}{20} = 0.5, \quad Gini = 0.5, \quad H = 1.0$$



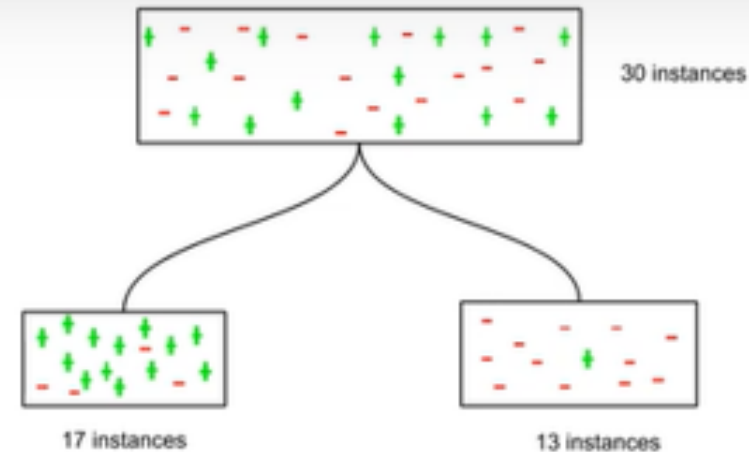
The reduction in Gini impurity due to a split

Gini impurity of a split:

$$G(S) = \frac{|S_L|}{|S|} G(S_L) + \frac{|S_R|}{|S|} G(S_R)$$

Where,

- $(S = S_L \cup S_R)$
- $S_L \cap S_R = \emptyset$
- $\frac{|S_L|}{|S|}$, fraction of inputs in left subtree
- $\frac{|S_R|}{|S|}$, fraction of inputs in right subtree
- The weighing allows giving more weight to larger subsets



$$\frac{17}{30} G(S_L) + \frac{13}{30} G(S_R)$$

Information Gain of a split

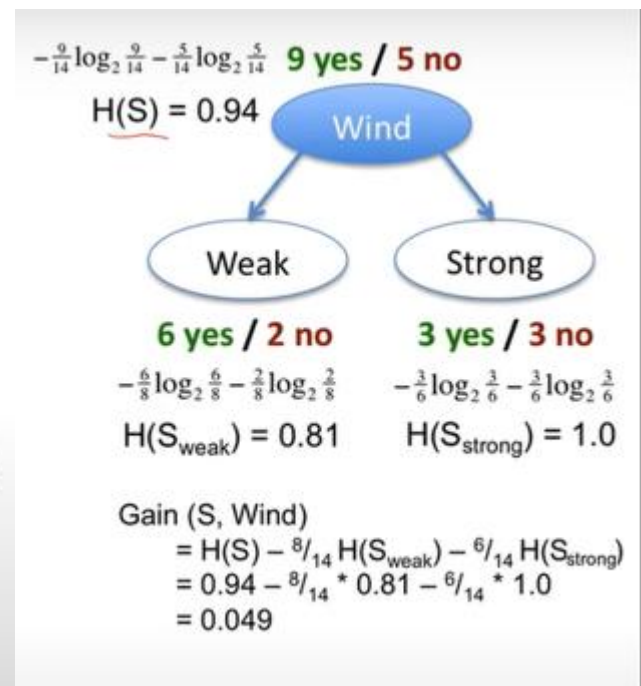
Information gain of a split:

Information Gain = Entropy(parent) - [weighted mean Entropy(children)]

$$\text{Gain}(S, A) = H(S) - \left(\frac{|S_L|}{|S|} H(S_L) + \frac{|S_R|}{|S|} H(S_R) \right)$$

Where,

- A is the attribute being used in the split
- $(S = S_L \cup S_R)$
- $S_L \cap S_R = \emptyset$
- $\frac{|S_L|}{|S|}$, fraction of inputs in left subtree
- $\frac{|S_R|}{|S|}$, fraction of inputs in right subtree
- The weighing allows giving more weight to larger subsets
- Cycle through splits using all unused features
- Choose the feature that **maximizes** the gain
- This is also the **Mutual Information** between the attribute A and the class labels of S



Measuring Impurity for Regression

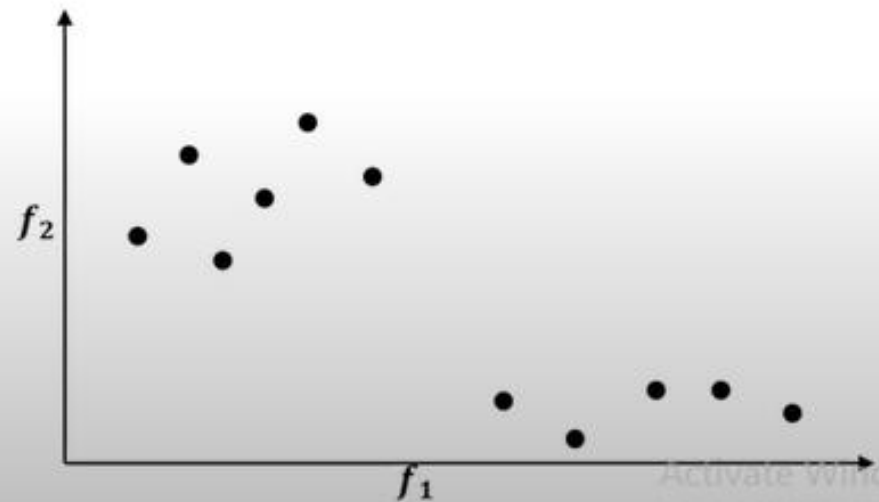
- Average squared difference from average label (variance).
 - For the set S :

$$L(S) = \frac{1}{|S|} \sum_{(x,y) \in S} (y - \bar{y}_S)^2$$

$$\text{where } \bar{y}_S = \frac{1}{|S|} \sum_{(x,y) \in S} y$$

- Weighted mean variance of a split:

$$L(S) = \frac{|S_L|}{|S|} L(S_L) + \frac{|S_R|}{|S|} L(S_R)$$



Thank you!

